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- Model Checking and Assessment
- Posterior predictive checking
  - Marginal Predictive Checking
  - Sensitivity analysis
  - Example

Bayesian Statistics and Data Analysis
Lecture 8a

Måns Magnusson
Department of Statistics, Uppsala University
Thanks to Aki Vehtari, Aalto University



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Mini-project comments

- Describe the priors as part of the model.



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Mini-project comments

- Describe the priors as part of the model.
- Summarise computational reliability in the report. Put detailed diagnostics in an appendix.



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- Describe the priors as part of the model.
- Summarise computational reliability in the report. Put detailed diagnostics in an appendix.
- Make figure labels and legends readable at the report's text size.
- Report length: 3.5–4 pages, excluding references and appendices.



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Mini-project comments

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- Summarise computational reliability in the report. Put detailed diagnostics in an appendix.
- Make figure labels and legends readable at the report's text size.
- Report length: 3.5–4 pages, excluding references and appendices.
- For `posteriordb`: describe the data or simulation method. Use the paper's exact model and priors for this replication study.



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Section 1

Model Checking and Assessment



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Model checking and predictive performance

- L8a: Model checking
  What features of the data does our model fail to reproduce?



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Model checking and predictive performance

- L8a: Model checking
  What features of the data does our model fail to reproduce?
- L8b: Predictive performance
  How well does it predict the observations we care about?



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Model checking and predictive performance

- L8a: Model checking
  What features of the data does our model fail to reproduce?
- L8b: Predictive performance
  How well does it predict the observations we care about?
- Winning a comparison does not establish adequate fit.
  Even the best candidate model may miss important features of the data.



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The Box process: Probabilistic modeling

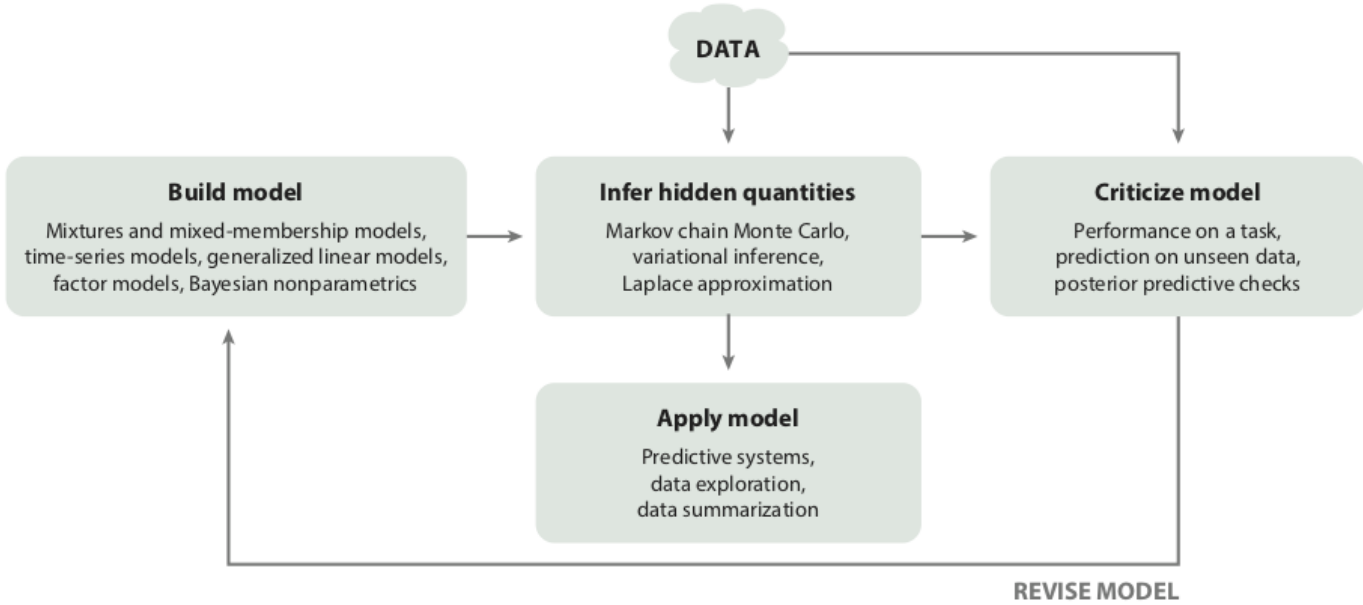


Figure: The Box approach (Box, 1976, Blei, 2014)



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Model assessment

- Sensibility with respect to additional information not used in model
  - e.g., if posterior would claim that hazardous chemical decreases probability of death



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Model assessment

- Sensibility with respect to additional information not used in model
  - e.g., if posterior would claim that hazardous chemical decreases probability of death
- External validation
  - compare predictions to completely new observations



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Model assessment

- Sensibility with respect to additional information not used in model
  - e.g., if posterior would claim that hazardous chemical decreases probability of death
- External validation
  - compare predictions to completely new observations
- Internal validation
  - posterior predictive checking
  - cross-validation predictive checking



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Section 2

Posterior predictive checking



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Posterior predictive checking – example

- Newcomb's speed of light measurements ($n = 66$)
  - $y_i \mid \mu, \sigma \sim \mathcal{N}(\mu, \sigma^2)$, conditionally independently
  - improper prior $p(\mu, \log \sigma) \propto 1$, or $p(\mu, \sigma) \propto 1/\sigma$



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- Generate one replicated dataset $y^{\text{rep},(s)}$:



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  - repeat both steps for $s = 1, \ldots, S$: S datasets of size n

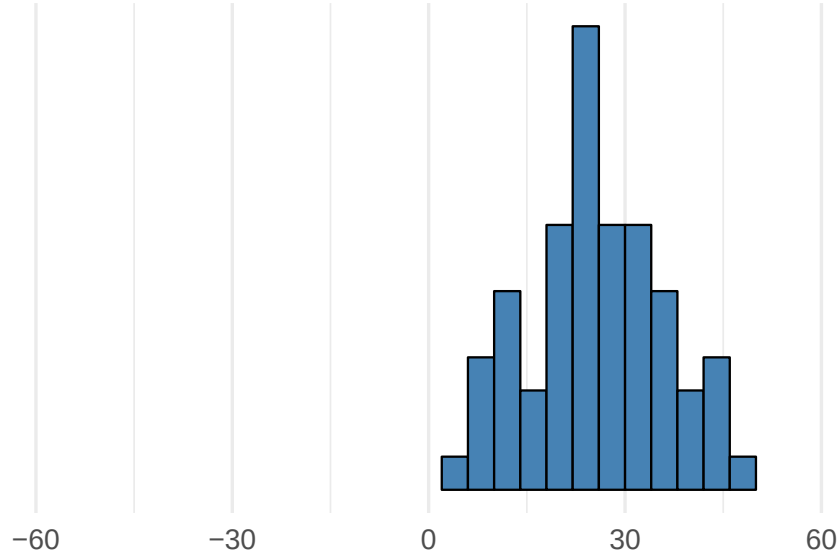


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- Generate one replicated dataset $y^{\text{rep},(s)}$:
  - draw one pair $(\mu^{(s)}, \sigma^{(s)})$ from $p(\mu, \sigma \mid y)$
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  - repeat both steps for $s = 1, \ldots, S$: S datasets of size n
  - one replicated dataset:





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Replicates vs. predictive distributions

- $\tilde{y}$ denotes an unobserved outcome or dataset to be predicted.



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Replicates vs. predictive distributions

- $\tilde{y}$ denotes an unobserved outcome or dataset to be predicted.
- y^{rep} denotes a replication of the whole experiment: the same sample size and experimental design, including fixed covariates.
- Both are drawn from predictive distributions. For a replicated dataset,

$$p(y^{\text{rep}} \mid y) = \int p(y^{\text{rep}} \mid \theta) p(\theta \mid y) d\theta.$$

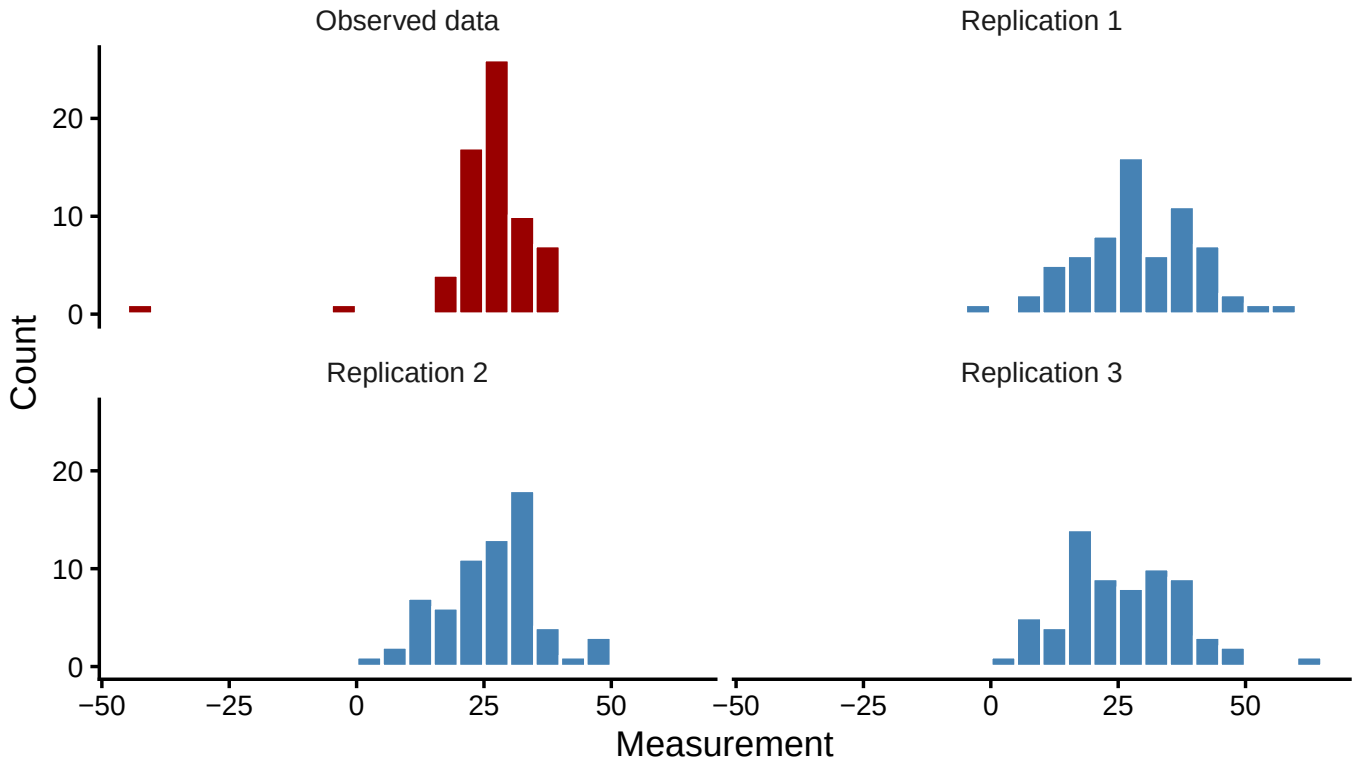


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Newcomb: observed and replicated data

Compare datasets of the same size, using the same axes.





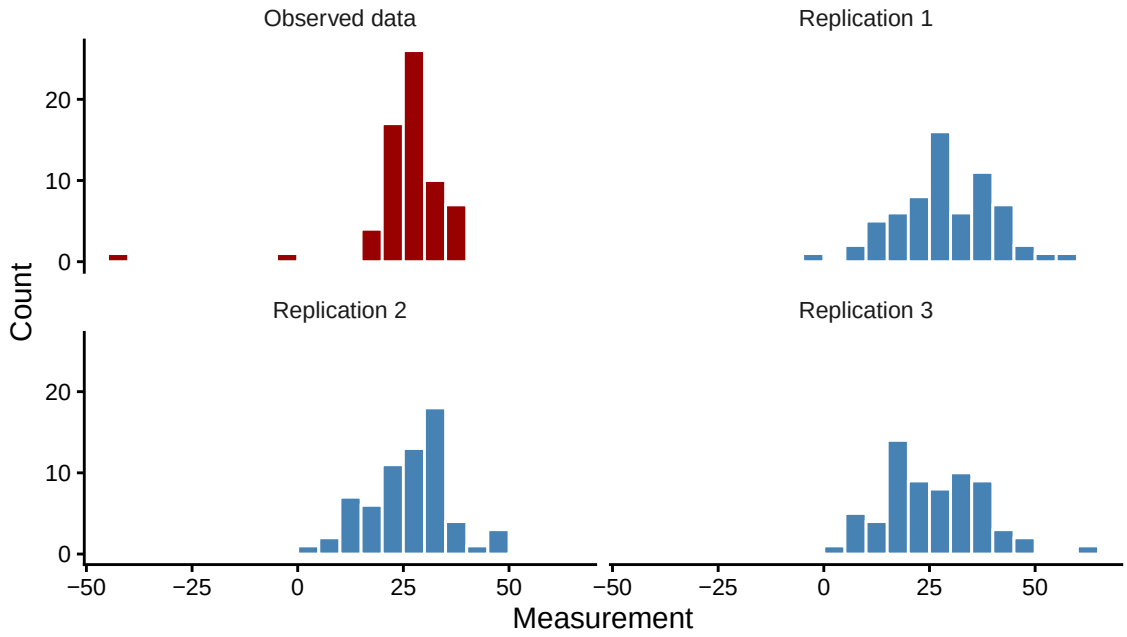
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Pair discussion: Newcomb's data

Discuss in pairs (1–2 minutes).

Which feature of the observed data is poorly reproduced? What statistic would you use to check it?





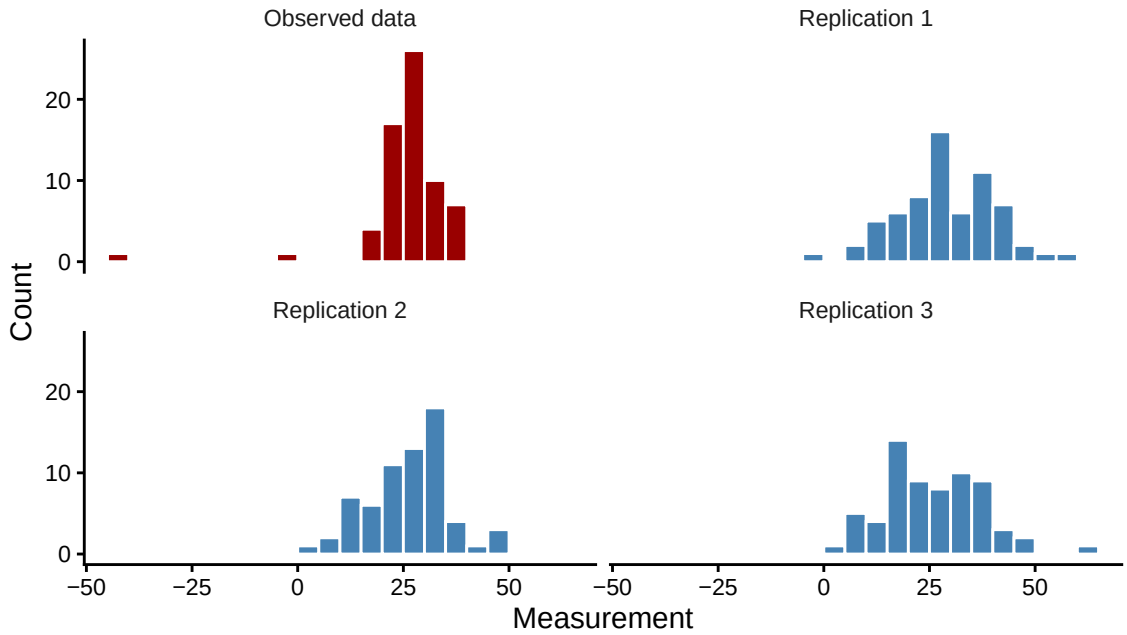
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Pair discussion: Newcomb's data

Discuss in pairs (1–2 minutes).

Which feature of the observed data is poorly reproduced? What statistic would you use to check it?



Answer: The unusually low observations are poorly reproduced. A useful check is $T(y) = \min_i y_i$, compared with the minima of *many* replicated datasets.



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Posterior predictive checking with test statistic

- Replicated data sets y^{rep}
- Test quantity (or discrepancy measure) $T(y, \theta)$
  - summary quantity for the observed data $T(y, \theta)$
  - summary quantity for a replicated data $T(y^{\text{rep}}, \theta)$
  - can be easier to compare summary quantities (y^{rep} statistics) than data sets



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Posterior predictive checking – example

- Compute test statistic for data $T(y, \theta) = \min(y)$



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Posterior predictive checking – example

- Compute test statistic for data $T(y, \theta) = \min(y)$
- Compute test statistic $\min(y^{\text{rep}})$ for many replicated datasets

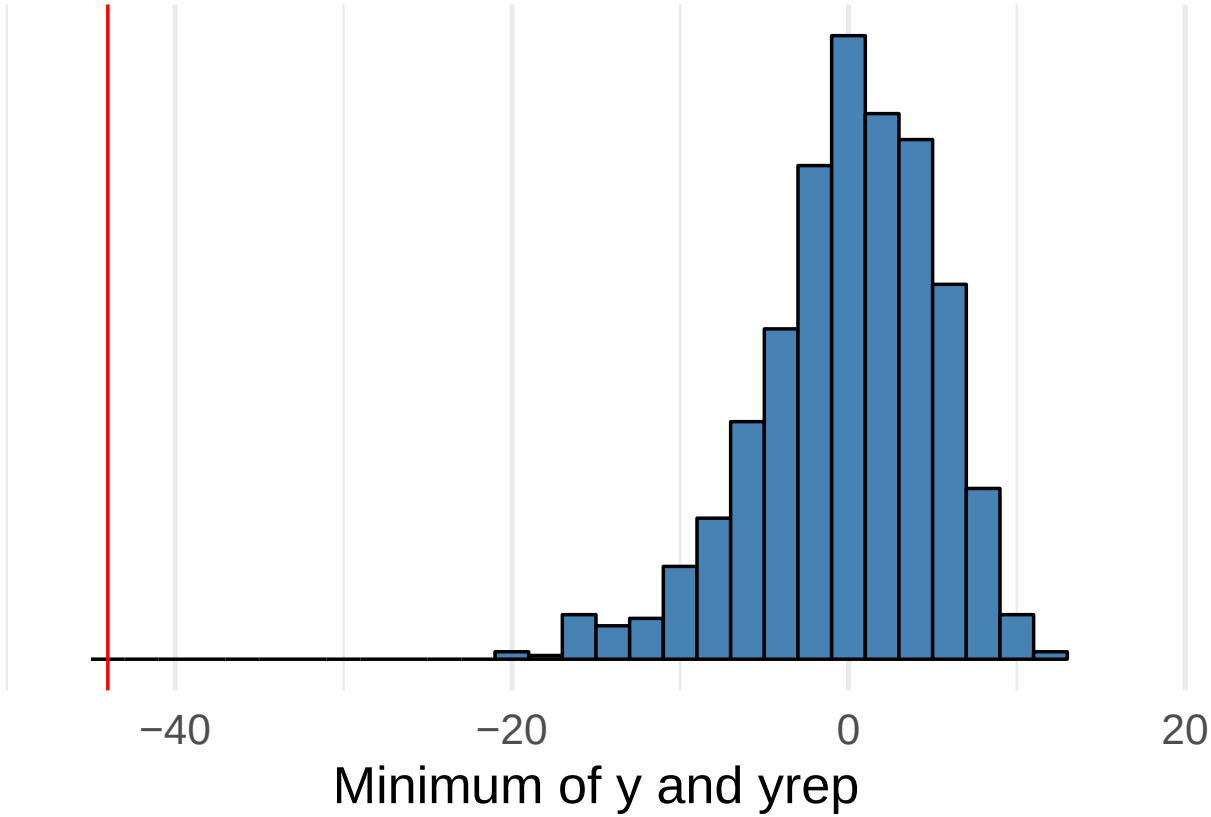


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Choosing a discrepancy: ancillarity

- A statistic is ancillary if its sampling distribution does not depend on the unknown parameters.



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Choosing a discrepancy: ancillarity

- A statistic is ancillary if its sampling distribution does not depend on the unknown parameters.
- Example: in a Normal model with known σ^2, the sampling distribution of s^2 does not depend on the unknown μ. If σ^2 is also unknown, s^2 is no longer ancillary.



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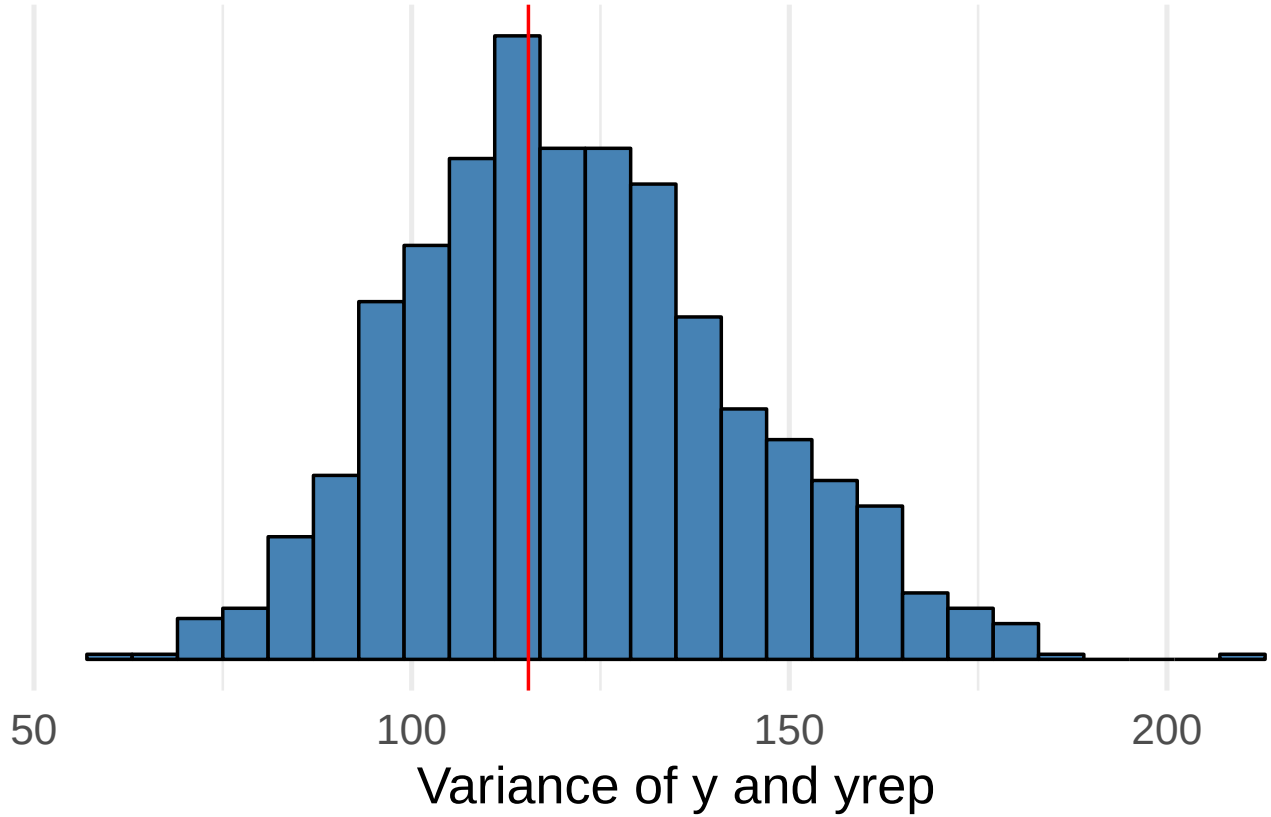
- A statistic is ancillary if its sampling distribution does not depend on the unknown parameters.
- Example: in a Normal model with known σ^2, the sampling distribution of s^2 does not depend on the unknown μ. If σ^2 is also unknown, s^2 is no longer ancillary.
- For checking, target features the model may miss, such as tails, skewness or dependence. A useful discrepancy need not be ancillary.



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Posterior predictive checking – a less informative check





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Pair discussion: a reassuring variance?

Discuss in pairs (1–2 minutes).

The observed variance looks typical among the variances of the replicated datasets.

Does this mean that the Normal model fits well? Why, or why not?



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Pair discussion: a reassuring variance?

Discuss in pairs (1–2 minutes).

The observed variance looks typical among the variances of the replicated datasets.

Does this mean that the Normal model fits well? Why, or why not?

Answer: No. The model uses the observed variance to learn σ^2, so reproducing that feature is a weak check.

A model can reproduce the variance while missing the shape of the distribution, including its lower tail. We need checks of several relevant features.



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Posterior predictive p-value

- How often is the replicated discrepancy at least as large?

$$p_B = \Pr(T(y^{\text{rep}}, \theta) \geq T(y, \theta) \mid y)$$
$$= \iint I\{T(y^{\text{rep}}, \theta) \geq T(y, \theta)\} p(y^{\text{rep}} \mid \theta) p(\theta \mid y) dy^{\text{rep}} d\theta.$$



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$$\widehat{p}_B = \frac{1}{S} \sum_{s=1}^{S} I\{T(y^{\text{rep},(s)}, \theta^{(s)}) \geq T(y, \theta^{(s)})\}.$$

Here $I\{\cdot\}$ is 1 if the condition holds, and 0 otherwise.



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- Values near 0 or 1 can reveal a mismatch; a non-extreme value does not establish that the model fits well.



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  Here $I\{\cdot\}$ is 1 if the condition holds, and 0 otherwise.
- Values near 0 or 1 can reveal a mismatch; a non-extreme value does not establish that the model fits well.
- Not the probability that the model is correct; not generally uniform under a correct model.



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Subsection 1

Marginal Predictive Checking



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Marginal and CV predictive checking

- For continuous observations, consider each predictive distribution:

$$p_i = \Pr(y_i^{\text{rep}} \leq y_i \mid y).$$

The observation y_i is also used to fit the model. These tail areas are not generally uniform, even for a correct model.



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Marginal and CV predictive checking

- For continuous observations, consider each predictive distribution:

$$p_i = \Pr(y_i^{\text{rep}} \leq y_i \mid y).$$

  The observation y_i is also used to fit the model. These tail areas are not generally uniform, even for a correct model.
- Leave-one-out probability integral transform (LOO-PIT):

$$u_i = \Pr(\tilde{y}_i \leq y_i \mid y_{-i}) = F_{-i}(y_i).$$

  Here F_{-i} is the predictive CDF fitted without y_i.
- For calibrated continuous predictions, approximate uniformity on $[0, 1]$ is a useful benchmark, not an exact small-sample guarantee.



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Marginal predictive checking - Example

- Posterior predictive tail areas (the same data are used for fitting):

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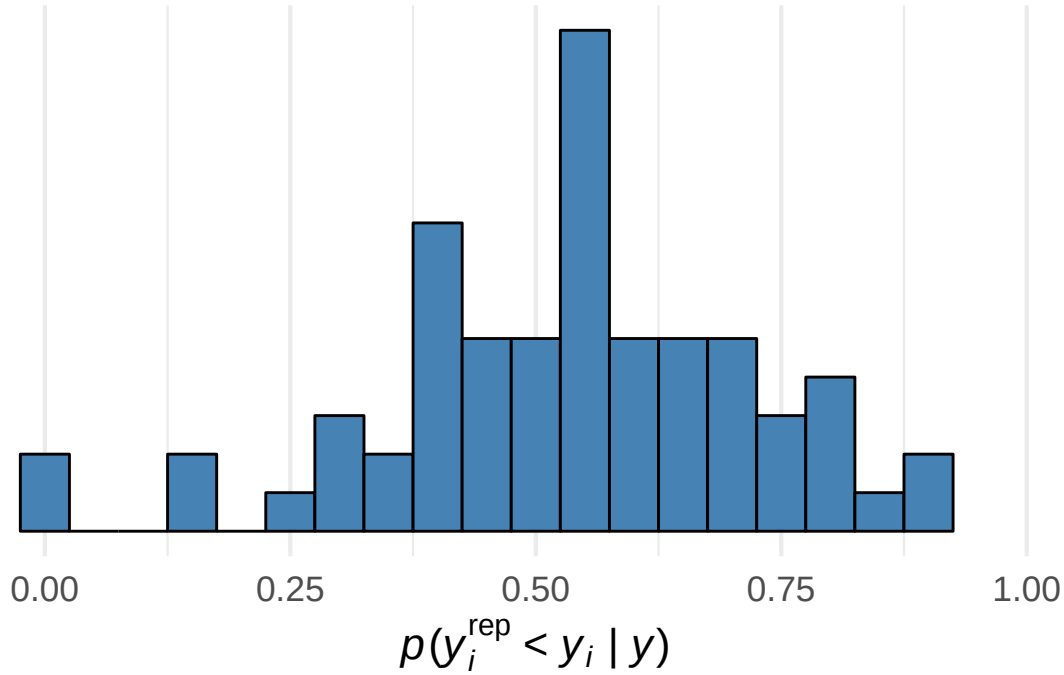
- Uniformity is not an exact reference for this plot. LOO-PIT reduces the reuse of observations in fitting and checking.

Marginal predictive checking - Example

- Posterior predictive tail areas (the same data are used for fitting):

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Subsection 2

Sensitivity analysis



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Sensitivity analysis

- How much different choices in model structure and priors affect the results



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Sensitivity analysis

- How much different choices in model structure and priors affect the results
  - test different models and priors



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Sensitivity analysis

- How much different choices in model structure and priors affect the results
  - test different models and priors
  - alternatively combine different models to one model
    - e.g. hierarchical model instead of separate and pooled
    - e.g. Student's t approaches a Gaussian as $\nu \to \infty$
  - robust models are good for testing sensitivity to "outliers"
    - e.g. t instead of Gaussian



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  - robust models are good for testing sensitivity to "outliers"
    - e.g. t instead of Gaussian
- Compare sensitivity of essential inference quantities
  - extreme quantiles are often more sensitive than means and medians
  - extrapolation is often more sensitive than interpolation



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Subsection 3

Example



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Example: Exposure to air pollution

- Example from Jonah Gabry, Daniel Simpson, Aki Vehtari, Michael Betancourt, and Andrew Gelman (2019). Visualization in Bayesian workflow. https://doi.org/10.1111/rssa.12378
- Estimation of human exposure to air pollution from particulate matter measuring less than 2.5 microns in diameter ($\mathrm{PM}_{2.5}$)
  - Exposure to $\mathrm{PM}_{2.5}$ is linked to a number of poor health outcomes and a recent report estimated that $\mathrm{PM}_{2.5}$ is responsible for three million deaths worldwide each year (Shaddick et al., 2017)
  - In order to estimate the public health effect of ambient $\mathrm{PM}_{2.5}$, we need a good estimate of the $\mathrm{PM}_{2.5}$ concentration at the same spatial resolution as our population estimates.

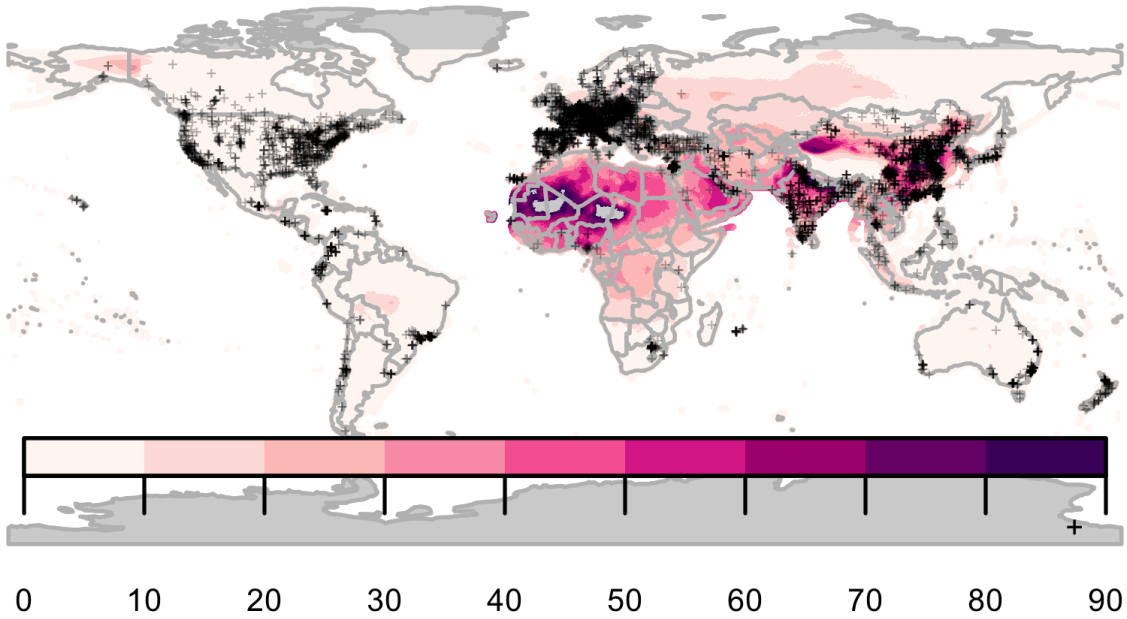


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Example: Exposure to air pollution

- Direct measurements of PM 2.5 from ground monitors at 2980 locations
- High-resolution satellite data of aerosol optical depth



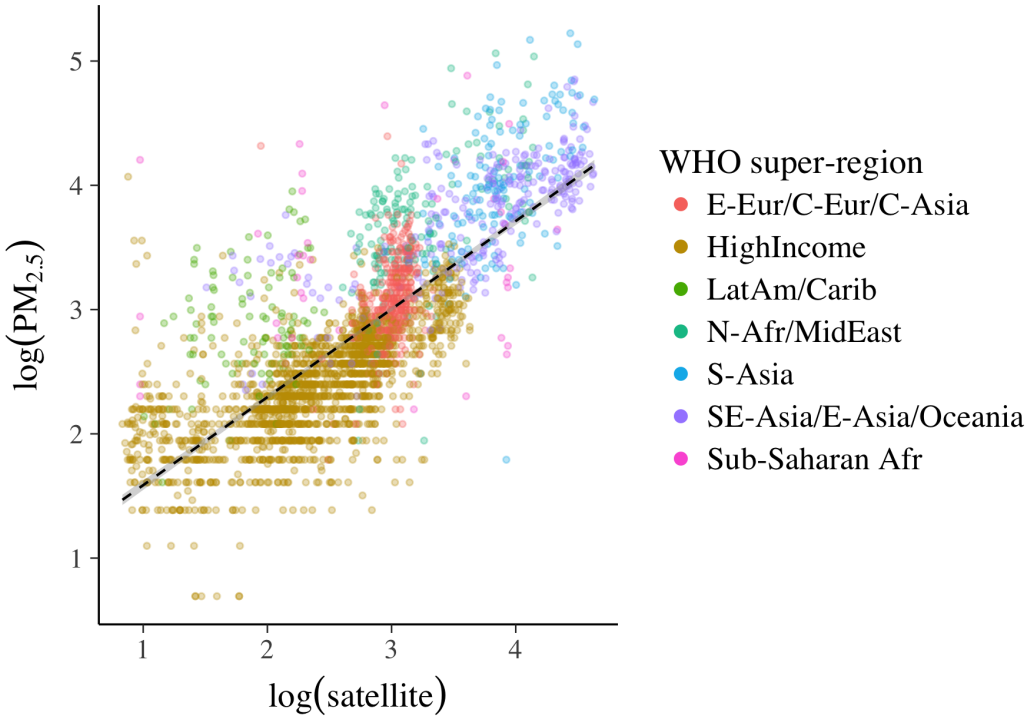


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- Model Checking and Assessment
- Posterior predictive checking
  - Marginal Predictive Checking
  - Sensitivity analysis
  - Example

Example: Exposure to air pollution

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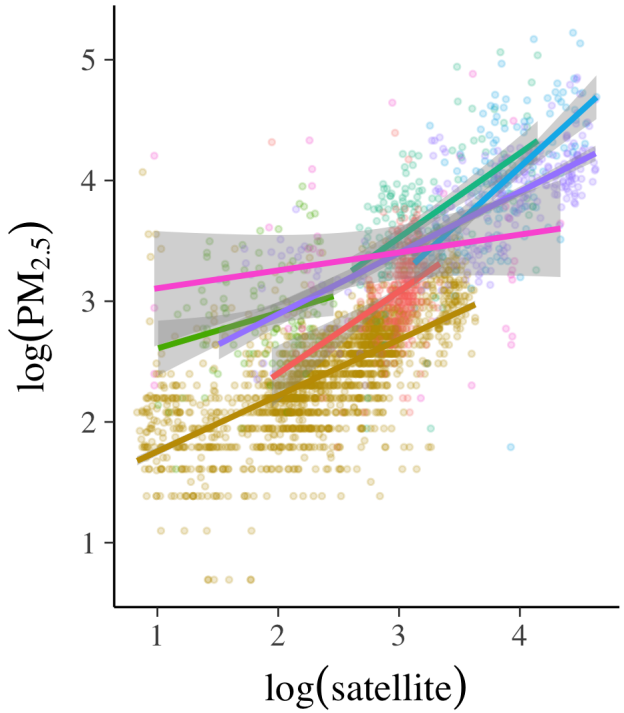


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Example: Exposure to air pollution

- Three models:
  1. Linear regression

$$y_{ij} \sim N(\beta_0 + \beta_1 x_{ij}, \sigma^2)$$

  2. Hierarchical linear regression (WHO super regions)
  3. Hierarchical linear regression (clustered super regions)

$$y_{ij} \sim N(\beta_0 + \beta_{0j} + (\beta_1 + \beta_{1j})x_{ij}, \sigma^2)$$

$$\beta_{0j} \sim N(0, \tau_0^2), \qquad \beta_{1j} \sim N(0, \tau_1^2)$$

Here y_{ij} is the natural log of $PM_{2.5}$ concentration, x_{ij} is the natural log of the satellite estimate, and j indexes regions. The model also requires priors for $\beta_0, \beta_1, \tau_0, \tau_1, \sigma$.



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Air pollution: which priors change?

Both concentrations are measured in $\mu\mathrm{g}/\mathrm{m}^3$. We model their natural logarithms: $y = \ln(c)$, $x = \ln(s)$.

Vague priors

$$\beta_0, \beta_1 \sim N(0, 100^2),$$
$$\tau_0^2, \tau_1^2, \sigma^2 \sim \mathrm{IG}(1, 100).$$

Weakly informative

$$\beta_0 \sim N(0, 1),$$
$$\beta_1 \sim N(1, 1),$$
$$\tau_0, \tau_1, \sigma \sim N_+(0, 1).$$

N uses variance as its second parameter. IG is inverse-gamma with shape and scale; N_+ is half-normal on positive values.

The tighter priors centre the calibration around intercept 0 and slope 1.

Priors follow the simulation code accompanying Gabry et al. (2019).

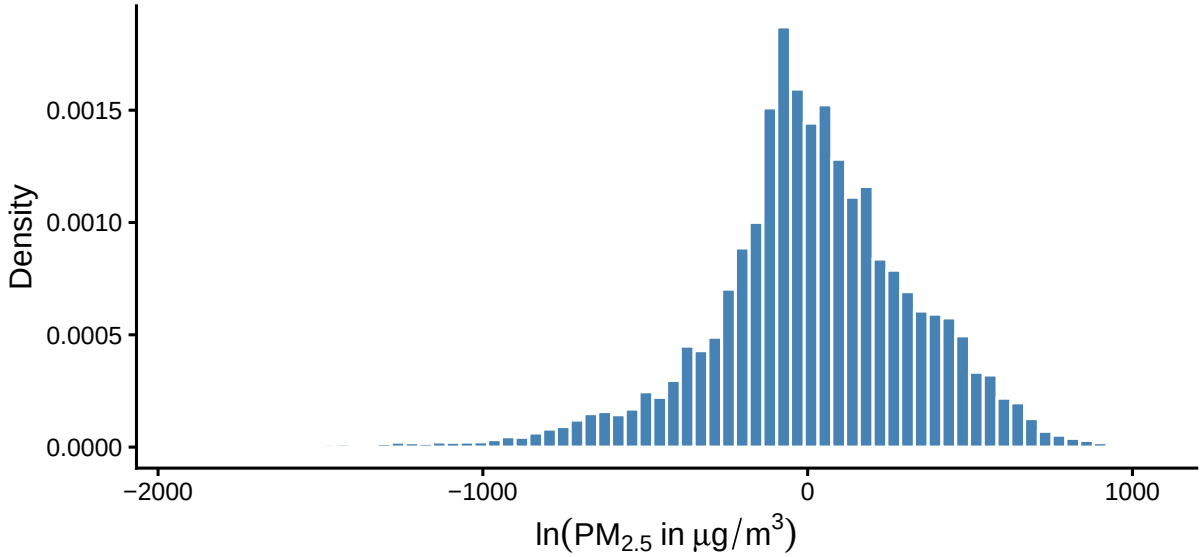


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Air pollution: prior predictive check

Vague priors



100 datasets at the observed design. Both plots use ln, with different horizontal scales.

$y = 0,\ 2.30,\ 4.61$ means approximately $c = 1,\ 10,\ 100\ \mu\mathrm{g/m}^3$.

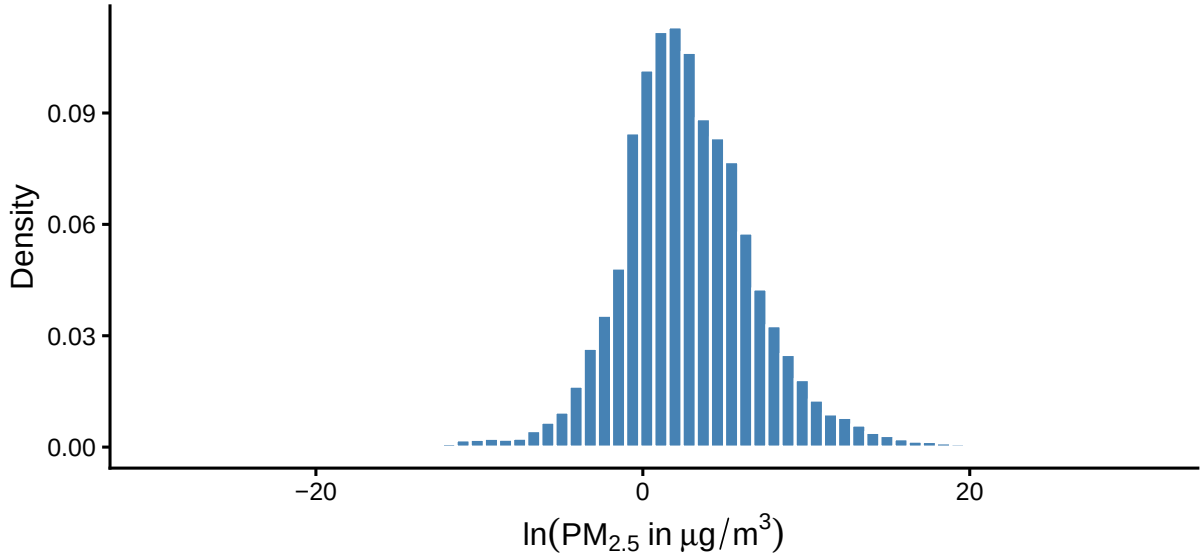


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Air pollution: prior predictive check

Weakly informative priors



100 datasets at the observed design. Both plots use ln, with different horizontal scales.

$y = 0,\ 2.30,\ 4.61$ means approximately $c = 1,\ 10,\ 100\ \mu\mathrm{g/m}^3$.

Even the tighter priors allow extreme concentrations.



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Example: Marginal predictive distributions

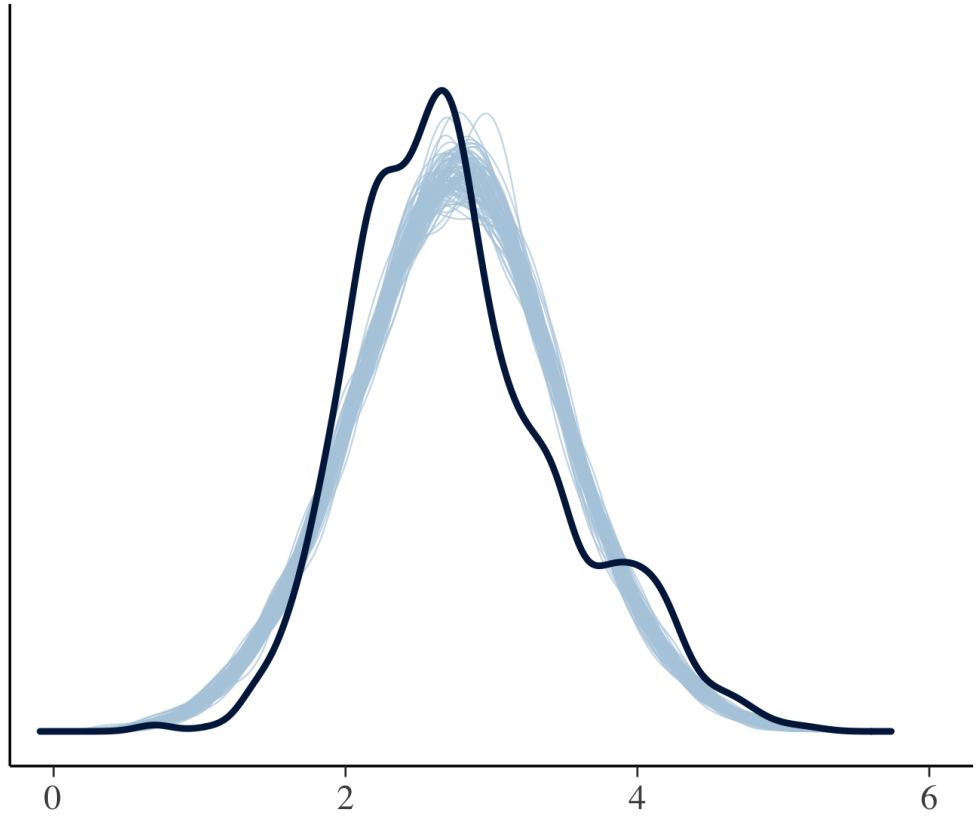


Figure: Model 1

Kernel density estimate of the observed dataset y (dark curve), with density estimates for 100 simulated datasets $y^{(\text{rep})}$ drawn from the posterior predictive distribution (thin, lighter lines)



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Example: Marginal predictive distributions

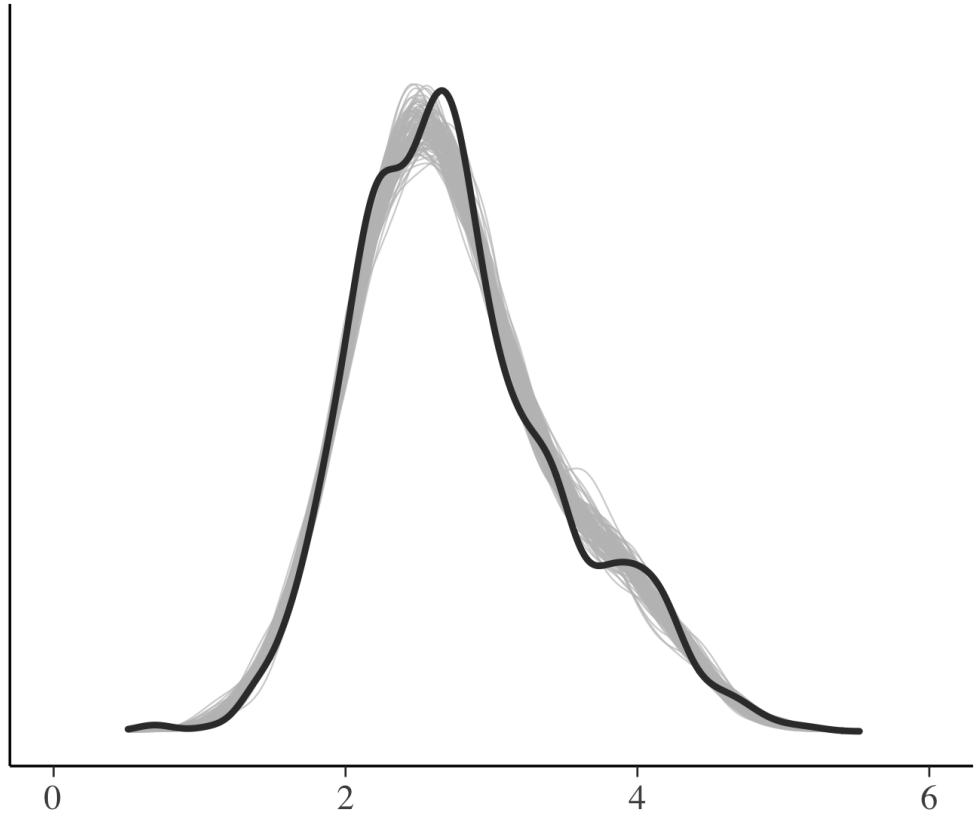


Figure: Model 2

Kernel density estimate of the observed dataset y (dark curve), with density estimates for 100 simulated datasets $y^{(\text{rep})}$ drawn from the posterior predictive distribution (thin, lighter lines)



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Example: Marginal predictive distributions

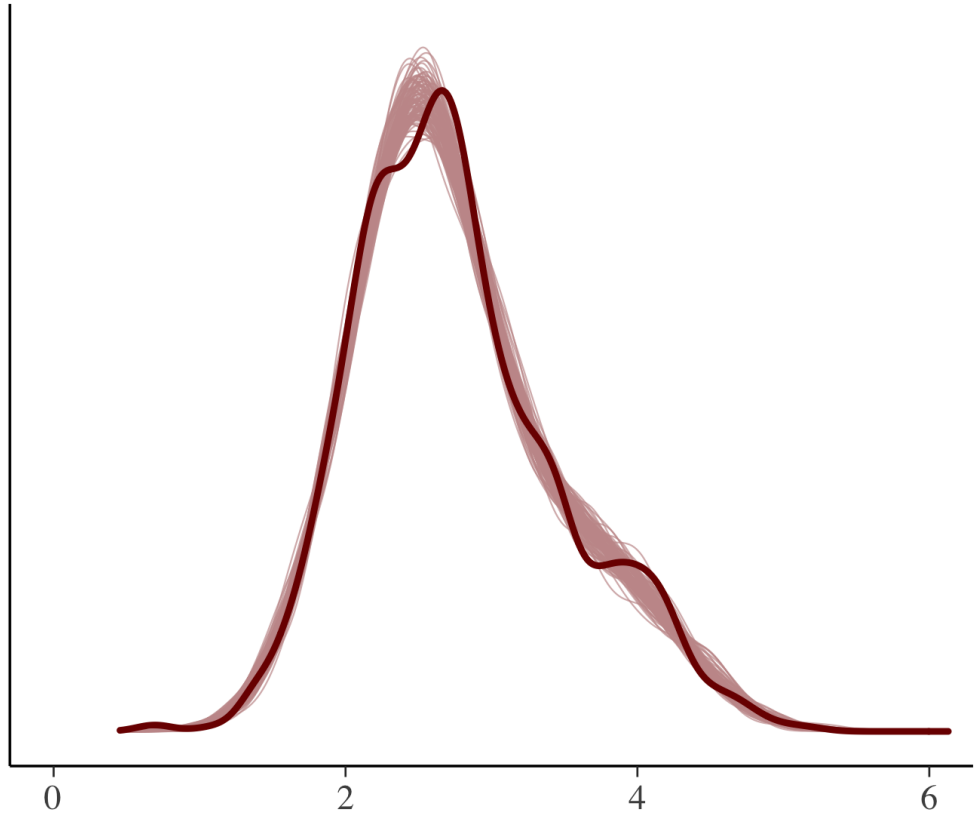


Figure: Model 3

Kernel density estimate of the observed dataset y (dark curve), with density estimates for 100 simulated datasets $y^{(\text{rep})}$ drawn from the posterior predictive distribution (thin, lighter lines)



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- Model Checking and Assessment
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Example: Test statistic (skewness)

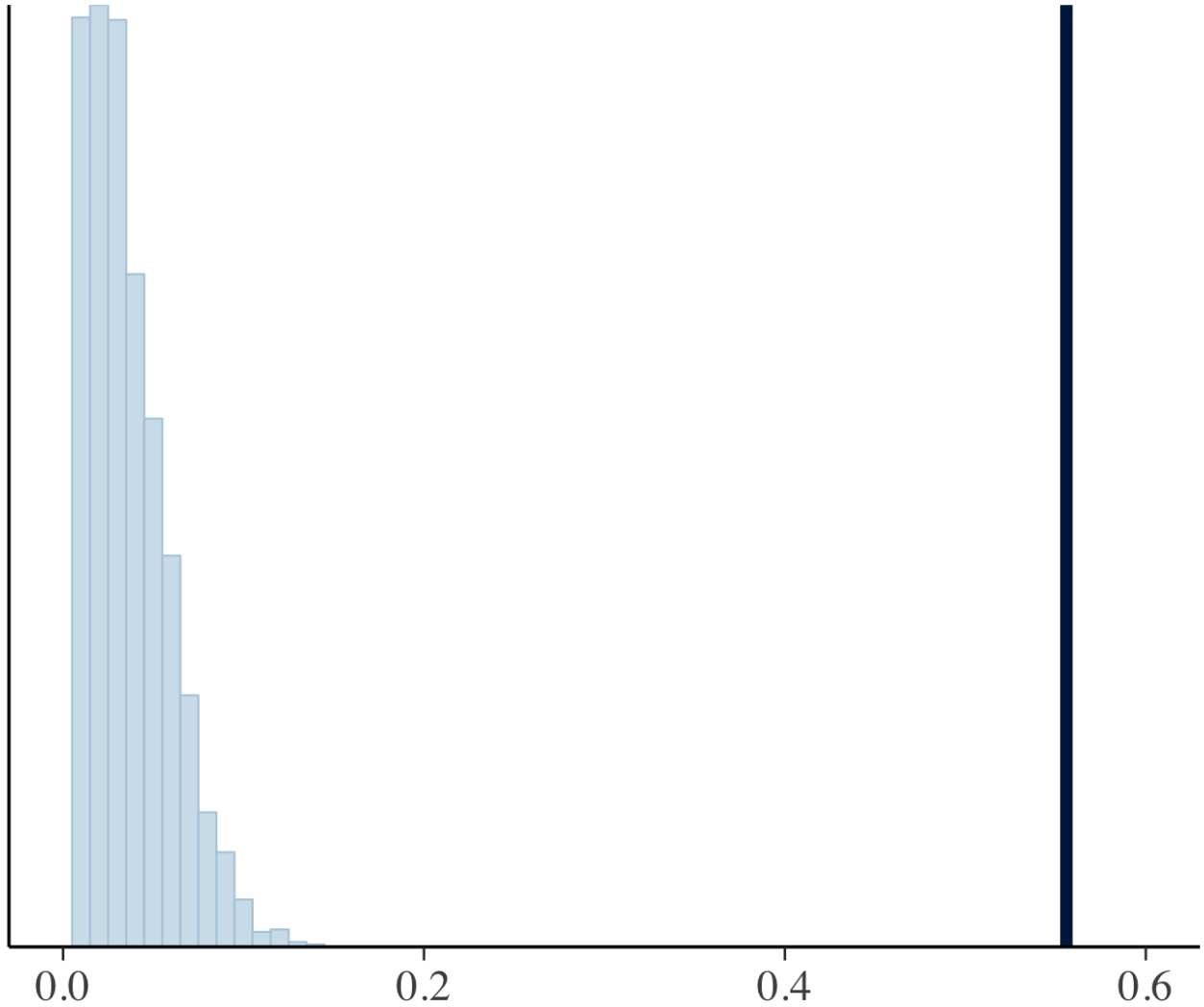


Figure: Model 1



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Example: Test statistic (skewness)

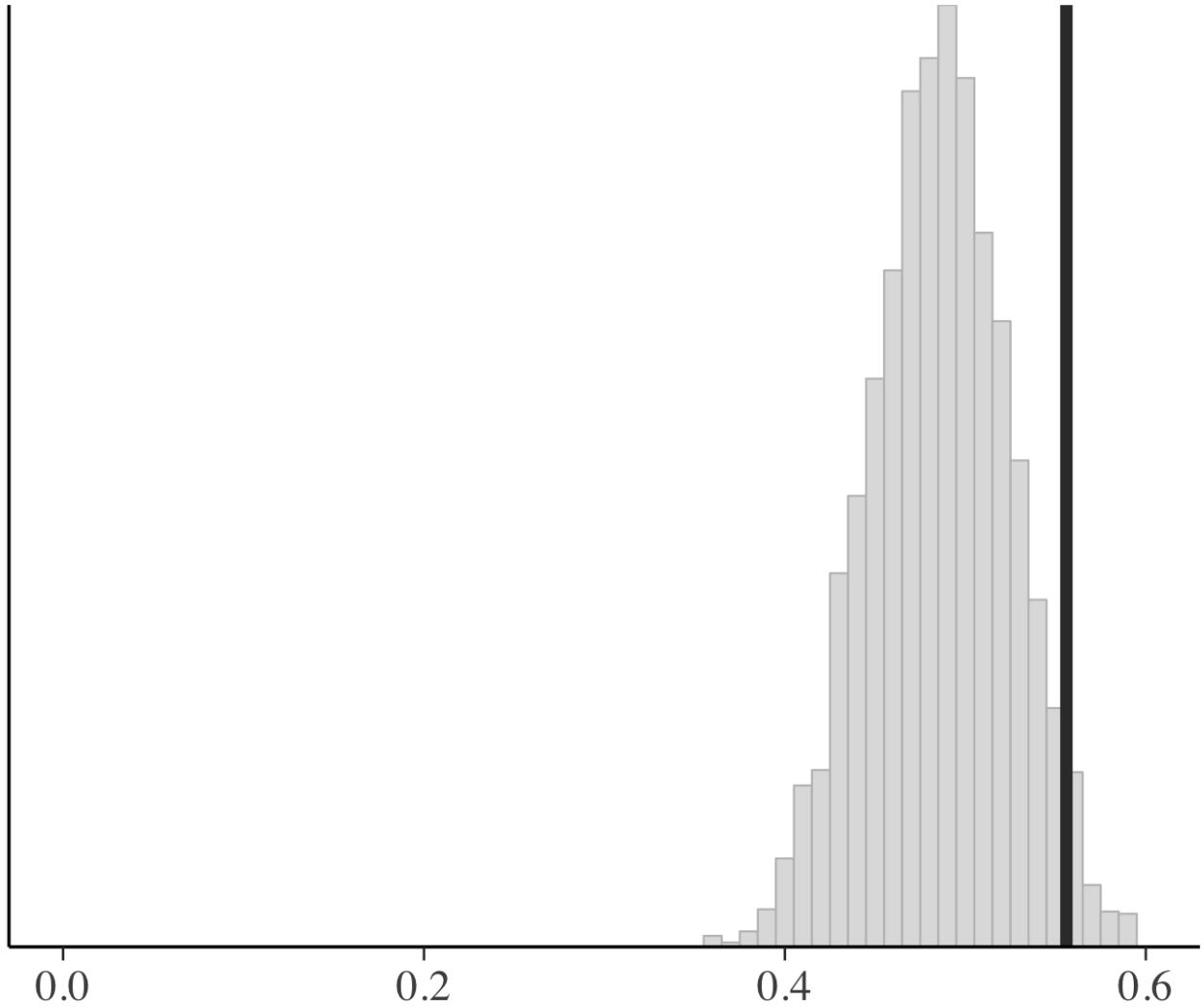


Figure: Model 2



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Example: Test statistic (skewness)

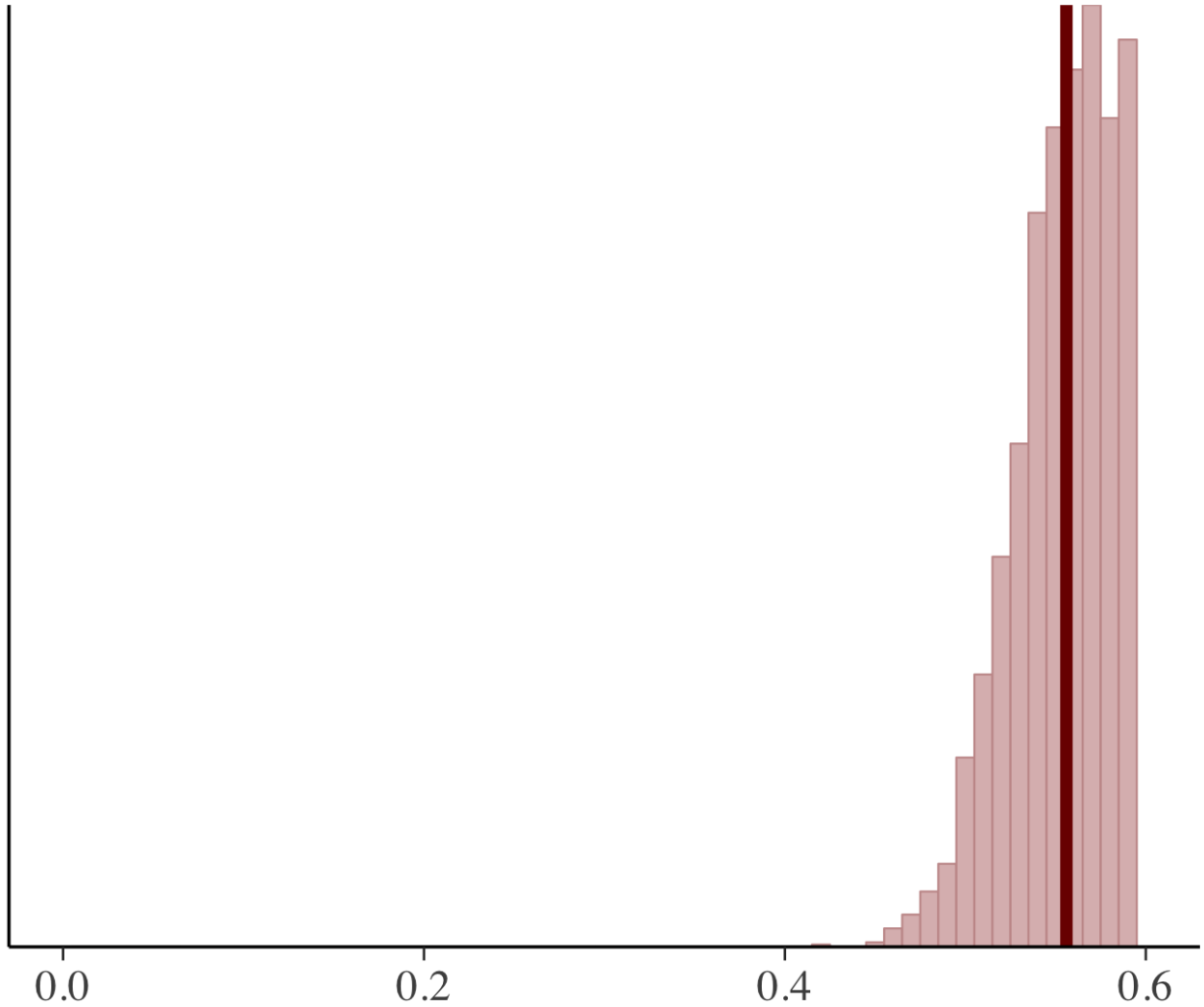


Figure: Model 3